Finding Thurstone: modeling comparative judgment data with R (and Stan)

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Abstract

The classical CJ analysis has become the standard approach for analyzing comparative judgment (CJ) data in education, linguistics, and psychology. Researchers favor it because it offers a simple method for measuring traits and conducting follow-up analyses. Its simplicity rests on two key features: the reliance on narrowly specified Thurstone-based models to estimate latent traits, and the use of ad hoc procedures for subsequent analyses. Recent studies, however, question whether the assumptions of these models–particularly the Bradley-Terry-Luce (BTL) model–hold in contemporary CJ applications, and whether the ad hoc procedures effectively fulfill their intended analytical goals.

To address these concerns, Rivera et al. (2025) proposed the ITCJ analysis, an umbrella analytical workflow designed to accommodate the complexity often observed in applied CJ settings. The approach enables the development of models tailored to the assumed data-generating process of the CJ system under study, thereby reducing reliance on simplifying assumptions. Moreover, by integrating measurement and inference within a single framework, it also removes dependence on ad hoc analytical procedures. Nonetheless, these potential advantages require empirical validation.

Thus, this study validates the ITCJ analysis using a simulated speech-quality dataset, benchmarking its performance against the classical CJ analysis. It also demonstrates its practical implementation, adopting a tutorial format with step-by-step guidance on data simulation, prior specification, model estimation, and interpretation using `R`, `Stan`, `cmdstan`, and `brms`. Ultimately, the study equips researchers with practical tools to apply the model to more complex CJ assessments.

Keywords: tutorial, causal inference, bayesian inference, thustone, comparative judgement

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1. Introduction

Comparative judgment (CJ) has emerged as a valuable methodology for measuring latent traits across diverse fields, including education (Kimbell, 2012; Jones and Inglis, 2015; van Daal et al., 2016; Bartholomew et al., 2018), political sciences (ZuccoJr. et al., 2019), linguistics (Boonen et al., 2020), and criminology (Seymour et al., 2022; Seymour and Hernandez, 2025). In CJ studies, judges actively compare pairs of stimuli to determine which stimulus exhibits more of the latent trait of interest (Thurstone, 1927b,a).

A specific data analysis workflow has become the standard approach for analyzing CJ data in education, linguistics, and psychology (see, e.g., Wu, 2025; Thwaites and Paquot, 2024; Maydeu-Olivares and Böckenholt, 2005). In this study, we refer to this workflow as the classical CJ analysis (hereafter, *CCJ analysis*). Researchers favor this approach because it provides a convenient and simple method for measuring traits and conducting follow-up analyses (Andrich, 1978; Pollitt, 2012b). This simplicity rests on two key features. First, it relies on narrowly specified Thurstone-based models to estimate latent traits. These models facilitate trait estimation by imposing several simplifying assumptions about traits, judges, and stimuli present in the CJ assessments under study (Thurstone, 1927a; Bramley, 2008). Second, the approach uses ad hoc procedures to conduct subsequent analyses, including data summarization and statistical inference (Pollitt, 2012b).

Recent studies, however, question whether the assumptions of these models–particularly the Bradley-Terry-Luce (BTL) model (Bradley and Terry, 1952; Luce, 1959)–hold in contemporary CJ applications, and whether the ad hoc procedures achieve their intended analytical goals (Bramley, 2008; Kelly et al., 2022; Rivera et al., 2025). For instance, Rivera et al. (2025) note that while BTL model assumptions of equal dispersions and zero correlations between stimuli simplify trait measurement, they may fail to represent complex traits or heterogeneous stimuli adequately (Thurstone, 1927b; Andrich, 1978; van Daal et al., 2016; Lesterhuis et al., 2018; Chambers and Cunningham, 2022). Thus, such assumptions can compromise the reliability and accuracy of trait estimates (Ackerman, 1989; Zimmerman, 1994; McElreath, 2020; Wu et al., 2022; Miller, 2023; Hoyle, 2023). Moreover, although ad hoc procedures simplify data analyses, the use of untested methods can also undermine the validity of inferences derived from CJ data (McElreath, 2020; Kline, 2023; Hoyle, 2023).

To address these concerns, Rivera et al. (2025) proposed the Information-Theoretical analysis for CJ (hereafter, *ITCJ analysis*), an umbrella analytical workflow designed to accommodate the complexity often observed in applied CJ settings. The approach leverages causal and Bayesian inference methods to combine Thurstone's core theoretical principles with key design features of CJ assessment. This integration enables researchers to develop models tailored to the assumed data-generating process of the CJ system under study, thereby reducing reliance on simplifying

assumptions. Moreover, by integrating measurement and inference within a single framework, the ITCJ analysis also eliminates the need for ad hoc analytical procedures.

The flexibility of the ITCJ analysis is particularly valuable in education, linguistics, and psychology, where CJ data often arises from complex assessment designs (Bramley, 2008; van Daal et al., 2017; Kelly et al., 2022). Its relevance, however, extends well beyond these fields to any discipline implementing CJ assessments, as the approach enables researchers to model the processes they seek to understand rather than those constrained by restrictive statistical models.

1.1. Research goals

The ITCJ analysis (Rivera et al., 2025) shows theoretical promise for producing reliable trait estimates and accurate statistical inferences. However, as the authors acknowledge, this promise has not yet been empirically validated. This study addresses this gap by pursuing two closely related research goals. The first goal is to *empirically validate* the proposed ITCJ analysis using a simulated speech-quality dataset. This validation assesses the accuracy and reliability of its trait estimates and inference parameters, benchmarked against the CCJ analysis.

The second goal emerges as a practical extension of this validation: *to demonstrate its practical implementation.* To this end, the document adopts a tutorial format, providing step-by-step guidance on data simulation, prior specification, model estimation, and interpretation using `R` (R Core Team, 2015), `Stan`(Stan Development Team., 2026b,a), and their interface packages `cmdstanr` (Gabry et al., 2025) and `brms` (Bürkner, 2017, 2018). Through this demonstration, the study equips researchers with the tools needed to apply the approach to more complex CJ assessments.

The remainder of this manuscript is organized into five sections. Section 2 reviews the primary analytical approach applied to CJ data: the CCJ analysis, and compares it with the ITCJ analysis. Section 3 describes the assumed data-generating process for the simulated dataset, the simulation procedure, the practical implementation of each analytical approach, and the evaluation criteria aligned with the research goals. Section 4 presents the data description and modeling results. Section 5 interprets the findings, outlines future research directions, and considers the study limitations. Finally, Section 6 offers the concluding remarks.

2. A tale of two analytical approaches

Pairwise comparison data can be analyzed using two main approaches: the CCJ and the ITCJ analysis. This section provides a detailed description of both.

2.1. The CCJ analysis

In practice, the CCJ analysis proceeds through a sequence of distinct analytical steps, each serving a distinct purpose (Pollitt, 2012a,b; Jones et al., 2019; Boonen et al., 2020; Chambers and Cunningham,

2022; Bouwer et al., 2023). The approach typically includes one or more of the following steps. First, researchers fit narrowly specified Thurstone-based models to the CJ data, yielding two primary outputs: (1) point estimates of stimulus latent scores with their associated standard errors, and (2) residuals at the comparison level. These outputs serve as the foundation for any follow-up analyses.

Second, researchers summarize or fit models to the estimated mean latent scores. This step serves multiple purposes, including aggregating stimulus-level point estimates to the individual level, partitioning variability between and within individuals, and drawing inferences about factors that influence trait scores. For example, Maydeu-Olivares and Böckenholt (2005) fitted multiple factor models to latent means that represent individuals' preferences for compact cars, aiming to provide a more parsimonious representation of these preferences. Similarly, Boonen et al. (2020) applied a multilevel regression model to stimulus mean latent scores to examine whether children's age or hearing status affects their intelligibility scores.

A third step may involve summarizing or modeling the residuals. Researchers typically use this step to aggregate the remaining variability at the judge level, partition residual variability between and within judges, test for systematic biases, and identify potential misfitting judgments, stimuli, or judges. For instance, Wu (2025) fitted an analysis of variance (ANOVA) model to judges' infit statistic to examine whether judge expertise influences judgment behavior. The infit statistics–a weighted average of the squared Pearson residuals–potentially indicate the extent to which judges' decisions deviate from expectations relative to other judges (Wright and Masters, 1982).

The CCJ analysis has become standard practice in education, linguistics, psychology, and even criminology (see e.g., Wu, 2025; Thwaites and Paquot, 2024; Maydeu-Olivares and Böckenholt, 2005; Seymour and Hernandez, 2025). Nevertheless, despite its widespread adoption, this stepwise approach has two important limitations: it treats outputs from earlier steps as fixed quantities, and it imposes restrictive models at each step.

Regarding the first limitation, the CCJ analysis treats outputs from earlier stages as fixed quantities rather than as uncertain parameter estimates, thereby limiting the propagation of uncertainty from one step to the next (Pritikin, 2020). When researchers ignore this uncertainty, estimates may become biased and less precise. The direction and magnitude of these biases can be unpredictable: effects may be attenuated, inflated, or remain unaffected depending on the degree of uncertainty in the scores and the true effects under investigation (McElreath, 2020; Kline, 2023; Hoyle, 2023). Moreover, reduced precision lowers statistical power and increases the risk of Type I or Type II errors (McElreath, 2020).

Regarding the second limitation, researchers often impose restrictive models at each stage of the CCJ analysis, which limit their ability to capture key features of the data-generating process. Although recent work incorporates selected features, most studies propose new Thurstone-based

models that address only one or two at a time. As a result, important data components remain unrepresented, leading to model misspecification and potentially biased and imprecise parameter estimates (Everitt and Skrondal, 2010).

For example, Seymour and Hernandez (2025) extended the BTL model to include tied outcomes and spatially correlated stimuli. They applied the model to an honour-based abuse dataset. Despite the model sophistication, the authors did not assess potential bias arising from the convenience sample, including judge biases and judge-specific effects. Moreover, because the analysis was limited to a single ward within one county, they did not account for ward-level effects or hierarchical clustering structures, such as wards nested within counties. This oversight restricts the generalizability of the findings. Notably, similar limitations are evident in Pritikin (2020), Schramm and Batchelder (2021), Seymour et al. (2022), and Gray et al. (2024), where models omit relevant sources of heterogeneity or design features that may affect parameter estimates, and ultimately, inferences.

Notably, the ITCJ analysis directly addresses these two limitations by providing a unified and systematic framework for analyzing CJ data (Rivera et al., 2025). Specifically, it allows researchers to model the processes of substantive interest while propagating uncertainty across components, rather than relying on restrictive statistical model specifications.

2.2. The ITCJ analysis

The approach begins by explicitly representing the relationships among observed judgments, discriminal differences, stimulus traits, individual traits, judge biases, and the sampling and comparison mechanisms using a general *Directed Acyclic Graph (DAG)* and a corresponding *Structural Causal Model (SCM)* (Morgan and Winship, 2014; Gross et al., 2018; Neal, 2020). This general DAG and SCM establish a coherent theoretical foundation for analysis of CJ data. Next, the approach adapts the general SCM and DAG to the assumed data-generating process of the CJ system under study. It then derives one or more bespoke *probabilistic* and *statistical* models tailored to that system. Finally, it uses one or more of these models to estimate, summarize, or conduct statistical inference on the traits. Figure 1 illustrates an example of a general DAG structure.

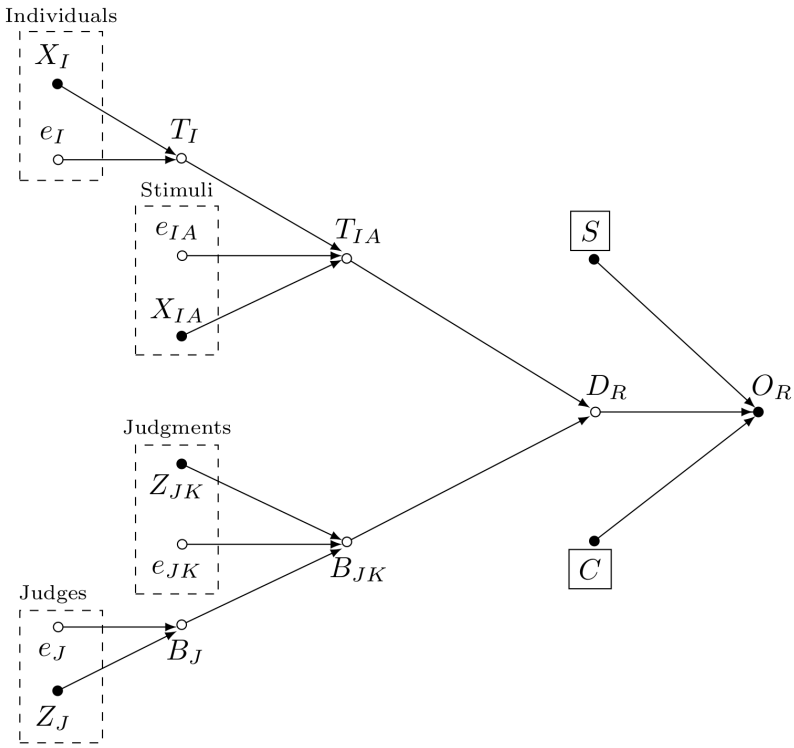

Figure 1: ITCJ analysis: DAG

In this representation, O_R denotes the observed judgment outcome vector, and D_R represents the discriminal difference vector. T_{IA} captures the vector of stimulus traits, and T_I represents the vector of individual traits (e.g., …). The vector of judgment biases is represented by B_{JK}, while the vector of judges' biases by B_J (e.g., …). Covariates at the stimulus and individual levels appear as X_{IA} and X_I, respectively; while Z_{JK} and Z_J represent covariates at the judgment and judge levels (these variables could be … , respectively). The error terms (e_{IA}, e_I, e_{JK}, e_J) capture residual variation at each level.

The DAG also represents the sampling and comparison mechanisms as the vectors S and C, two conditioned variables that determine how population outcomes become "observed" (sampled) outcomes. Importantly, the DAG depicts S and C as independent from all other variables by showing no arrows pointing into them. Regarding S, this indicates that the DAG applies to *Simple Random Sampling (SRSg)* designs, where each repeated judgment, judge, stimulus, and individual has an equal probability of inclusion within their respective groups (Lawson, 2015). Regarding C, the DAG applies to *Random Allocation Comparative Designs* (Bramley, 2015) or *Incomplete Block Designs* (Lawson, 2015), where every repeated judgment has an equal chance of being included in the sample.

Researchers can then translate this DAG and SCM representations into probabilistic forms that express the joint distribution of a complex CJ system as a product of simpler *conditional probability*

distributions (CPDs) (Pearl et al., 2016; Neal, 2020; see also Rivera et al., 2025, section 5), as illustrated in Figure 2.

$$
\begin{aligned}
O_R &:= f_O(D_R, S, C) \\
D_R &:= f_D(T_{IA}, B_{JK}) \\
T_{IA} &:= f_T(T_I, X_{IA}, e_{IA}) \\
T_I &:= f_T(X_I, e_I) \\
B_{JK} &:= f_B(B_J, Z_{JK}, e_{JK}) \\
B_J &:= f_B(Z_J, e_J) \\
\\
e_I &\perp \{e_J, e_{IA}, e_{JK}\} \\
e_J &\perp \{e_{IA}, e_{JK}\} \\
e_{IA} &\perp e_{JK}
\end{aligned}
$$

(a) SCM

$$
\begin{aligned}
&P(O_R \mid D_R, S, C) \\
&P(D_R \mid T_{IA}, B_{JK}) \\
&P(T_{IA} \mid T_I, X_{IA}, e_{IA}) \\
&P(T_I \mid X_I, e_I) \\
&P(B_{JK} \mid B_J, Z_{JK}, e_{JK}) \\
&P(B_J \mid Z_J, e_J) \\
\\
&P(e_I)P(e_{IA})P(e_J)P(e_{JK})
\end{aligned}
$$

(b) Probabilistic model

Figure 2: ITCJ analysis. SCM (left) and probabilistic (right) representations for DAG in Figure 1.

Critically, the approach allows tailoring this general structure to a specific CJ context, enabling development of parsimonious models that match the assumed data-generating process without imposing unnecessary constraints. For example, Rivera et al. (2025) exemplified a CJ assessment designed to evaluate the impact of different teaching methods on students' writing ability. In this case, the observed outcome was binary, so the model assumed O_R followed a Bernoulli distribution. The discriminal difference (D_R) was determined by the texts' discriminal processes (T_{IA}) and the judgment biases (B_{JK}). Student-level variables X_I, such as teaching method, were included, whereas text-level variables X_{IA} (e.g., text length) were not gathered. Similarly, judge-level variables Z_J, like judgment expertise, were incorporated, while judgment-level variables Z_{JK} (e.g., number of judgments per judge) were absent. Finally, the probabilistic assumptions for the idiosyncratic errors (e_I, e_{IA}, e_J, e_{JK}) were specifically made to resolve indeterminacies in *location*, *orientation*, and *scale* for the variables T_I, T_{IA}, B_J, B_{JK}, as required in latent variable models (Depaoli, 2021; de Ayala, 2009).

Lastly, researchers can derive one or more *bespoke* statistical models tailored to the CJ system of interest, as demonstrated in Rivera et al. (2025). At this stage, the ITCJ analysis differs fundamentally from the CBTL approach in how it handles parameter estimation and inference. Rather than fitting multiple separate models, the approach simultaneously estimates all parameters within a single coherent framework using *Bayesian inference*. This joint estimation accounts for uncertainty at all levels and enables direct inference about quantities of interest without relying on

post-hoc procedures (McElreath, 2020).

3. Methods

3.1. Step 1, from Theory to Design: Data-generating assumptions

3.2. Step 2, from Design to Data: Data simulation

3.3. Step 5, from Estimator and Sample to Estimate(s): The analysis aproaches

3.3.1. The CCJ analysis

3.3.2. The ITCJ analysis

3.3.2.1. Model 1.

3.3.2.2. Model 2.

3.3.2.3. Model 3.

3.3.2.4. Model 4.

3.3.2.5. Model 5.

3.3.2.6. Model 6.

3.4. Step 6, from Estimate(s) to Diagnostics and Posterior predictives: The evaluation criteria

4. Results

4.1. Data description

4.2. Data modeling

4.2.1. The CCJ analysis

4.2.2. The ITCJ analysis

4.2.2.1. Model 1.

4.2.2.2. Model 2.

4.2.2.3. Model 3.

4.2.2.4. Model 4.

4.2.2.5. Model 5.

4.2.2.6. Model 6.

4.2.2.7. Model comparison.

5. Discussion

5.1. Future research directions

5.2. Study limitations

6. Conclusion

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Licence: All the code that is original to this study and not attributed to any other authors is copyrighted by Jose Manuel Rivera Espejo and released under the new BSD-3-Clause license.

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7. Appendix

7.1. Appendix A: Stationarity, converge and mixing

7.2. Appendix B: Misfit observations

7.3. Appendix C: Sample size calculations

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